US Patent & Trademark Office
Patent Public Search | Text View

United States Patent Application Publication	20250282041
Kind Code	A1
Publication Date	September 11, 2025
Inventor(s)	Davis; Anthony et al.

POWER TOOL WITH COMBINED CHIP FOR WIRELESS COMMUNICATIONS AND POWER TOOL CONTROL

Abstract

Power tool devices described herein include a motor, an actuator configured to be actuated by a user, a plurality of power switching elements configured to drive the motor, a gate driver coupled to the plurality of power switching elements and configured to control the plurality of power switching elements, a first printed circuit board (PCB), an antenna, and a combined chip. The combined chip is located on the first PCB and is coupled to the actuator, the antenna, and the gate driver. The combined chip includes a memory and an electronic processor configured to determine that the actuator has been actuated, and provide, in response to determining that the actuator has been actuated, a signal to the gate driver, control the signal based on the motor position information, wirelessly transmit power tool device information to an external device, and wirelessly receive configuration information from the external device via the antenna.

Inventors: **Davis; Anthony (Brookfield, WI), Huber; Alexander (Menomonee Falls, WI), Obermann; Timothy R. (Waukesha, WI)**

Applicant: **MILWAUKEE ELECTRIC TOOL CORPORATION** (Brookfield, WI)

Family ID: **73550152**

Appl. No.: **19/217388**

Filed: **May 23, 2025**

Related U.S. Application Data

parent US continuation 18661961 20240513 parent-grant-document US 12311522 child US 19217388
parent US continuation 18164907 20230206 parent-grant-document US 11986942 child US 18661961
parent US continuation 16879245 20200520 parent-grant-document US 11571803 child US

18164907
us-provisional-application US 62854656 20190530

Publication Classification

Int. Cl.: **B25F5/02** (20060101); **B25B21/02** (20060101); **B25D16/00** (20060101); **G01D5/14** (20060101); **H04M1/72412** (20210101); **H04M1/72415** (20210101)

U.S. Cl.:

CPC **B25F5/02** (20130101); **G01D5/145** (20130101); **H04M1/72412** (20210101); **H04M1/72415** (20210101); B25B21/02 (20130101); B25D16/00 (20130101); B25D2250/221 (20130101)

Background/Summary

RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/661,961, filed May 13, 2024, which is a continuation of U.S. patent application Ser. No. 18/164,907, filed Feb. 6, 2023, now U.S. Pat. No. 11,986,942, which is a continuation of U.S. patent application Ser. No. 16/879,245, filed May 20, 2020, now U.S. Pat. No. 11,571,803, which claims the benefit of U.S. Provisional Patent Application No. 62/854,656, filed on May 30, 2019, the entire content of each of which is hereby incorporated by reference.

FIELD
[0002] Embodiments described herein relate to power tools that communicate with an external device.
SUMMARY
[0003] Power tools described herein include a motor, and an actuator configured to be actuated by a user. The power tools further include a Hall effect sensor configured to monitor motor position information, and a plurality of power switching elements configured to drive the motor. The power tools further include a gate driver coupled to the plurality of power switching elements and configured to control the plurality of power switching elements. The power tools further include a printed circuit board (PCB), and a combined chip located on the PCB and coupled to the actuator, the Hall sensor, and the gate driver. The combined chip includes a memory, an integrated antenna, and an electronic processor. The electronic processor is configured to determine that the actuator has been actuated, and in response to determining that the actuator has been actuated, provide a signal to the gate driver. The gate driver is configured to control the plurality of power switching elements based on the signal. The electronic processor is further configured to receive the motor position information from the Hall effect sensor, and control the signal provided to the gate driver based on the motor position information. The electronic processor is further configured to transmit power tool device information to an external device via the integrated antenna, and receive configuration information from the external device via the integrated antenna. The electronic processor is configured to use the configuration information to determine the signal that is provided to the gate driver.
[0004] In some embodiments, the power tool further includes a second chip that is separate from the combined chip, and the second chip may include the gate driver.
[0005] Methods described herein for operating a power tool device include determining, with an electronic processor of the power tool device, that an actuator of the power tool device has been actuated by a user. The electronic processor is included in a combined chip that includes a memory

and an integrated antenna. The combined chip is located on a first printed circuit board (PCB) and coupled to the actuator. The methods also include providing, with the electronic processor, in response to determining that the actuator has been actuated, a signal to a gate driver. The gate driver is configured to control a plurality of power switching elements configured to drive a motor of the power tool device based on the signal. The methods also include receiving, with the electronic processor, motor position information of the motor from a Hall effect sensor, controlling, with the electronic processor, the signal provided to the gate driver based on the motor position information, wirelessly transmitting, with the electronic processor, power tool device information to an external device via the integrated antenna, wirelessly receiving, with the electronic processor, configuration information from the external device via the integrated antenna, and controlling, with the electronic processor, the signal that is provided to the gate driver based on the configuration information.

[0006] Power tool devices described herein include a motor, an actuator configured to be actuated by a user, a plurality of power switching elements configured to drive the motor, a gate driver coupled to the plurality of power switching elements and configured to control the plurality of power switching elements, a first printed circuit board (PCB), an antenna, and a combined chip. The combined chip is located on the first PCB and is coupled to the actuator, the antenna, and the gate driver. The combined chip includes a memory and an electronic processor configured to determine that the actuator has been actuated, and provide, in response to determining that the actuator has been actuated, a signal to the gate driver. The gate driver is configured to control the plurality of power switching elements based on the signal. The combined chip is also configured to determine motor position information, control the signal provided to the gate driver based on the motor position information, wirelessly transmit power tool device information to an external device via the antenna, and wirelessly receive configuration information from the external device via the antenna. The electronic processor is configured to use the configuration information to control the signal that is provided to the gate driver.

[0007] Before any embodiments are explained in detail, it is to be understood that the embodiments are not limited in its application to the details of the configuration and arrangement of components set forth in the following description or illustrated in the accompanying drawings. The embodiments are capable of being practiced or of being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein are for the purpose of description and should not be regarded as limiting. The use of “including,” “comprising,” or “having” and variations thereof are meant to encompass the items listed thereafter and equivalents thereof as well as additional items. Unless specified or limited otherwise, the terms “mounted,” “connected,” “supported,” and “coupled” and variations thereof are used broadly and encompass both direct and indirect mountings, connections, supports, and couplings.

[0008] In addition, it should be understood that embodiments may include hardware, software, and electronic components or modules that, for purposes of discussion, may be illustrated and described as if the majority of the components were implemented solely in hardware. However, one of ordinary skill in the art, and based on a reading of this detailed description, would recognize that, in at least one embodiment, the electronic-based aspects may be implemented in software (e.g., stored on non-transitory computer-readable medium) executable by one or more processing units, such as a microprocessor and/or application specific integrated circuits (“ASICs”). As such, it should be noted that a plurality of hardware and software based devices, as well as a plurality of different structural components, may be utilized to implement the embodiments. For example, “servers,” “computing devices,” “controllers,” “processors,” etc., described in the specification can include one or more processing units, one or more computer-readable medium modules, one or more input/output interfaces, and various connections (e.g., a system bus) connecting the components.

[0009] Relative terminology, such as, for example, “about,” “approximately,” “substantially,” etc., used in connection with a quantity or condition would be understood by those of ordinary skill to

be inclusive of the stated value and has the meaning dictated by the context (e.g., the term includes at least the degree of error associated with the measurement accuracy, tolerances [e.g., manufacturing, assembly, use, etc.] associated with the particular value, etc.). Such terminology should also be considered as disclosing the range defined by the absolute values of the two endpoints. For example, the expression “from about 2 to about 4” also discloses the range “from 2 to 4”. The relative terminology may refer to plus or minus a percentage (e.g., 1%, 5%, 10%, or more) of an indicated value.
[0010] It should be understood that although certain drawings illustrate hardware and software located within particular devices, these depictions are for illustrative purposes only. Functionality described herein as being performed by one component may be performed by multiple components in a distributed manner. Likewise, functionality performed by multiple components may be consolidated and performed by a single component. In some embodiments, the illustrated components may be combined or divided into separate software, firmware and/or hardware. For example, instead of being located within and performed by a single electronic processor, logic and processing may be distributed among multiple electronic processors. Regardless of how they are combined or divided, hardware and software components may be located on the same computing device or may be distributed among different computing devices connected by one or more networks or other suitable communication links. Similarly, a component described as performing particular functionality may also perform additional functionality not described herein. For example, a device or structure that is “configured” in a certain way is configured in at least that way but may also be configured in ways that are not explicitly listed.
[0011] Other aspects of the embodiments will become apparent by consideration of the detailed description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0012] FIG. **1** illustrates a communication system according to one example embodiment.
[0013] FIG. **2** illustrates a block diagram of an external device of the communication system of FIG. **1** according to one example embodiment.
[0014] FIGS. **3**A and **3**B illustrate an example control screen of a graphical user interface (GUI) on a touch screen display of the external device of FIGS. **1** and **2** according to one example embodiment.
[0015] FIG. **4** illustrates a power tool of the communication system of FIG. **1** according to one example embodiment.
[0016] FIGS. **5**A and **5**B illustrate simplified block diagrams of example prior art power tools.
[0017] FIG. **6** illustrates a simplified block diagram of the power tool of FIG. **4** according to one example embodiment.
[0018] FIG. **7** illustrates a simplified block diagram of the power tool of FIG. **4** according to another example embodiment.
[0019] FIG. **8** illustrates a further block diagram of the power tool of FIGS. **4** and **6** according to one example embodiment.
[0020] FIGS. **9**A, **9**B, and **9**C illustrate example locations within the power tool of FIGS. **4** and **6** where printed circuit boards (PCBs) may be positioned.
[0021] FIG. **10** illustrates a control PCB of the power tool of FIGS. **4** and **6** according to one example embodiment.
[0022] FIG. **11** illustrates a flowchart of a method performed by an electronic processor of a combined chip of the power tool of FIGS. **4**, **6**, and **8** according to one example embodiment.
DETAILED DESCRIPTION

[0023] FIG. **1** illustrates a communication system **100**. The communication system **100** includes power tool devices **102** and an external device **108**. Each power tool device **102** (e.g., battery powered impact driver **102***a*, power tool battery pack **102***b*, and mains-powered hammer drill **102***c*) and the external device **108** can communicate wirelessly while they are within a communication range of each other. Each power tool device **102** may communicate power tool device information such as power tool status, power tool operation statistics, power tool identification, stored power tool usage information, power tool maintenance data, and the like. Therefore, using the external device **108**, a user can access stored power tool usage or power tool maintenance data. With this tool data, a user can determine how the power tool device **102** has been used, whether maintenance is recommended or has been performed in the past, and identify malfunctioning components or other reasons for certain performance issues. The external device **108** can also transmit configuration information to the power tool device **102** for power tool configuration (e.g., motor control), firmware updates, or to send commands (e.g., turn on a work light). The configuration information from the external device **108** may also allow a user to set operational parameters, safety parameters, select tool modes, and the like for the power tool device **102**. In some embodiments, the power tool devices **102***a*, **102***b*, and **102***c* may communicate with each other (e.g., peer to peer communication to form a mesh network).

[0024] The external device **108** may be, for example, a smart phone (as illustrated), a laptop computer, a tablet computer, a personal digital assistant (PDA), or another electronic device capable of communicating wirelessly with the power tool device **102** and providing a user interface. The external device **108** generates the user interface and allows a user to access and interact with tool information. The external device **108** can receive user inputs to determine operational parameters, enable or disable features, and the like. The user interface of the external device **108** provides an easy-to-use interface for the user to control and customize operation of the power tool.

[0025] The external device **108** includes a communication interface that is compatible with a wireless communication interface or module of the power tool device **102**. The communication interface of the external device **108** may include a wireless communication controller (e.g., a Bluetooth® module), or a similar component. The external device **108**, therefore, grants the user access to data related to the power tool device **102**, and provides a user interface such that the user can interact with a processor of the power tool device **102**.

[0026] In addition, as shown in FIG. **1**, the external device **108** can also share the information obtained from the power tool device **102** with a remote server **112** connected by a network **114**. The remote server **112** may be used to store the data obtained from the external device **108**, provide additional functionality and services to the user, or a combination thereof. In one embodiment, storing the information on the remote server **112** allows a user to access the information from a plurality of different locations. In another embodiment, the remote server **112** may collect information from various users regarding their power tool devices and provide statistics or statistical measures to the user based on information obtained from the different power tools. For example, the remote server **112** may provide statistics regarding the experienced efficiency of the power tool device **102**, typical usage of the power tool device **102**, and other relevant characteristics and/or measures of the power tool device **102**. The network **114** may include various networking elements (routers, hubs, switches, cellular towers, wired connections, wireless connections, etc.) for connecting to, for example, the Internet, a cellular data network, a local network, or a combination thereof. In some embodiments, the power tool device **102** may be configured to communicate directly with the server **112** through an additional wireless communication interface or with the same wireless communication interface that the power tool device **102** uses to communicate with the external device **108**.

[0027] FIG. **2** illustrates a block diagram of the external device **108** of FIG. **1** according to one example embodiment. As shown in FIG. **2**, the external device **108** may include an electronic processor **205** (for example, a microprocessor or another electronic device). The electronic processor **205** may include input and output interfaces (not shown) and be electrically connected to a memory **210**, an external wireless communication controller **215**, and a touch screen display **220**. In some embodiments, the external device **108** may include fewer or additional components in configurations different from that illustrated in FIG. **2**. For example, in some embodiments, the external device **108** also includes a camera and a location determining component (for example, a global positioning system receiver).

[0028] The memory **210** includes read only memory (ROM), random access memory (RAM), other non-transitory computer-readable media, or a combination thereof. The electronic processor **205** is configured to receive instructions and data from the memory **210** and execute, among other things, the instructions. In particular, the electronic processor **205** executes instructions stored in the memory **210** to perform the functions of the external deice **108** described herein. In some embodiments, the memory **210** stores core application software, tool mode profiles, temporary configuration data, tool interfaces, tool data including received tool identifiers and received tool usage data (e.g., tool operational data).

[0029] The touch screen display **220** allows the external device **108** to output visual data to a user and receive user inputs. Although not illustrated, the external device **108** may include further user input devices (e.g., buttons, dials, toggle switches, and a microphone for voice control) and further user outputs (e.g., speakers and tactile feedback elements). Additionally, in some instances, the external device **108** has a display without touch screen input capability and receives user input via other input devices, such as buttons, dials, and toggle switches.

[0030] The external device **108** communicates wirelessly with other devices (e.g., the power tool devices **102***a*, **102***b*, and **102***c* and the server **112** of FIG. **1**) via the external wireless communication controller **215**, e.g., using a Bluetooth® or Wi-Fi® protocol. The external wireless communication controller **215** further communicates with the server **112** over the network **114**. The external wireless communication controller **215** includes at least one transceiver to enable wireless communications between the external device **108** and the power tool **104** and/or the server **112** through the network **114**. In some instances, the external wireless communication controller **215** includes two separate wireless communication controllers, one for communicating directly with the power tools devices **102** (e.g., using Bluetooth® or Wi-Fi® communications) and one for communicating through the network **114** (e.g., using Wi-Fi® or cellular communications).

[0031] In some embodiments, the server **112** includes an electronic processor, a memory, and an external wireless communication controller similar to the like-named components described above with respect to the external device **108**. These components may allow the server **112** to communicate with the external device **108** over the network **114**. The communication link between the server **112**, the network **114**, and the external device **108** may include various wired and wireless communication pathways, various network components, and various communication protocols.

[0032] Returning to the external device **108**, the electronic processor **205** may be configured to generate a graphical user interface (GUI) on the touch screen display **220** enabling the user to interact with the power tool **104** and server **112**. In some embodiments, a user may access a repository of software applications (e.g., an “app store” or “app marketplace”) using the external device **108** to locate and download core application software, which may be referred to as an “app.” In some embodiments, the app is obtained using other techniques, such as downloading from a website using a web browser on the external device **108**. As will become apparent from the description below, at least in some embodiments, the app on the external device **108** provides a user with the ability to control, access, and/or interact with a multitude of different tool features for a multitude of different types of tools.

[0033] FIGS. **3**A and **3**B illustrate an example control screen **305** of the GUI on the touch screen display **220** of the external device **108**. As shown in FIGS. **3**A and **3**B, the control screen **305** includes a top portion **305***a* and a bottom portion **305***b*. In some embodiments, the information shown on the control screen **305** and the features available to be configured by a user on the control screen **305** depend on a type of the power tool device **102**. As mentioned above, the external device **108** may receive user input via the touch screen display **220** and transmit configuration information to the power tool device **102** based on the received user input. In other words, the user is able to configure parameters of the power tool device **102** using the control screen **305**. For example, via the control screen **305**, the user is able to adjust a maximum speed of a motor of the power tool device **102** via a speed text box **310** or a speed slider **315**; enable/disable a custom drive control using a toggle **320**; alter a trigger ramp up parameter via slider **325** to adjust how quickly the motor ramps up to a desired speed upon trigger pull; adjust a work light duration with slider **330**, work light text box **335**, and “always on” toggle **340**; and adjust a work light intensity via the work light brightness options **345**. Upon enabling the toggle **320**, torque level control elements of the custom drive control become active and are no longer greyed-out, such that a user can adjust the torque level using a slider **350** or torque text box **355**.

[0034] The control screen **305** of FIGS. **3**A and **3**B is merely an example. In some embodiments, the external device **108** may be configured to display other configurable parameters of the power tool device **102** and send other configuration information to the power tool device **102** in response to receiving user inputs regarding the other configurable parameters. For example, the external device **108** may allow a user to set current or voltage limits and the like. As another example, the external device **108** may display a lockout parameter that allows the user to disable the power tool device **102** (e.g., prevent the motor from operating and/or prevent a work light from turning on) when a user actuates an actuator (i.e., trigger) to operate the motor and/or turn on the work light. In some embodiments, the lockout parameter may be set based on time such that the power tool device **102** locks out (i.e., is disabled) at a future time as set by a user. As another example, the external device **108** may display a geofencing parameter that allows the user to create a geofence such that when the power tool device **102** is moved outside of the geofence, the external device **108** provides a notification indicating that the power tool device **102** is outside of the geofence. As yet another example, the external device **108** may display a tracking feature that allows the user to determine a location at which the power tool device **102** is located.

[0035] Additionally, in some embodiments, the control screen **305** may display power tool device information received by the external device **108** from one or more power tool devices **102**. For example, the control screen **305** may display usage information and/or status information received from one or more power tool devices **102** by the external device **108** (e.g., an amount of time that the power tool device **102** has been in use, maintenance alerts, battery charge level, geofence boundary violations, and the like). As another example, the control screen **305** may display a list of nearby power tool devices **102** within communication range of the external device **108** and identification data (e.g., a number, name, and/or image) associated with each power tool device **102**.

[0036] Referring back to FIG. **1**, each power tool device **102** may be configured to perform one or more specific tasks (e.g., drilling, cutting, fastening, pressing, lubricant application, sanding, heating, grinding, bending, forming, impacting, polishing, lighting, etc.). For example, an impact wrench is associated with the task of generating a rotational output (e.g., to drive a bit), while a reciprocating saw is associated with the task of generating a reciprocating output motion (e.g., for pushing and pulling a saw blade). As another example, a dedicated work light is associated with the task of lighting a designated area such as a workspace. The task(s) associated with a particular power tool device may also be referred to as the primary function(s) of the power tool device. The particular power tool devices **102** illustrated and described herein (e.g., an impact driver) are merely representative. Other embodiments of the communication system **100** include a variety of types of power tool devices **102** (e.g., a power drill, a hammer drill, a pipe cutter, a sander, a nailer, a grease gun, etc.).

[0037] As an example of a power tool device **102**, FIG. **4** illustrates an impact driver **104** (herein power tool **104**). The power tool **104** is representative of various types of power tools that operate within system **100**. Accordingly, the description with respect to the power tool **104** in the system **100** is similarly applicable to other types of power tool devices such as other power tools, work lights, battery packs (e.g., power tool device **102***b* of FIG. **1**), and the like. As shown in FIG. **4**, the power tool **104** includes an upper main body **405**, a handle **410**, a battery pack receiving portion **415**, mode pad **420**, an output drive device or mechanism **425**, an actuator **430** (i.e., trigger), a work light **435**, and a forward/reverse selector **440**. The housing of the power tool **104** (e.g., the main body **405** and the handle **410**) are composed of a durable and light-weight plastic material. The drive device **425** may be composed of a metal (e.g., steel). The drive device **425** on the power tool **104** is a socket. However, each power tool **104** may have a different drive device **425** specifically designed for the task (or primary function) associated with the power tool **104**. For example, the drive device for a power drill may include a bit driver, while the drive device for a pipe cutter may include a blade. The battery pack receiving portion **415** is configured to receive and couple to the battery pack (e.g., **102***b* of FIG. **1**) that provides power to the power tool **104**. The battery pack receiving portion **415** includes a connecting structure to engage a mechanism that secures the battery pack and a terminal block to electrically connect the battery pack to the power tool **104**. The mode pad **420** allows a user to select a mode of the power tool **104** and indicates to the user the currently selected mode of the power tool **104**. In some embodiments, the modes selectable using the mode pad **420** are received by the power tool **104** from the external device **108** in response to user inputs setting different parameters via the control screen **305** (see FIGS. **3**A and **3**B).

[0038] With reference to FIGS. **5**A and **5**B that illustrate simplified block diagrams of example prior art power tools, some prior art power tools may include a motor **505**, a three-phase inverter **510** (including six power switching elements such as field-effect transistors (FETs)), a battery pack **515**, a power tool microcontroller **520**, a gate driver **525**, a power manager **530** for the gate driver **525**, a wireless communication microcontroller **535** (i.e., a Bluetooth® low energy (BLE) microcontroller), and a transceiver **540** (i.e., a BLE transceiver). The battery pack **515** may provide power to the power tool microcontroller **520** and the BLE microcontroller **535**. The power tool microcontroller **520** may control the gate driver **525** (e.g., by providing a pulse width modulation (PWM) signal based on actuation of the actuator **430**). In turn, the gate driver **525** may control the FETs of the three-phase inverter **510** to open/close to allow/disallow current from the battery pack **515** to be provided to coils of a stator of the motor **505** to cause a rotor of the motor **505** to rotate. The power manager **530** may monitor one or more characteristics of the three-phase inverter **510** and/or the motor **505** during operation (e.g., motor current) and may control the gate driver **525** based on the monitored characteristics. For example, the power manager **530** may control the gate driver **525** to prevent the motor **505** from operating in response to an over-current condition being detected. The power tool microcontroller **520** may receive configuration information (e.g., as described previously herein) from the external device **108** via the BLE microcontroller **535** and BLE transceiver **540**. Additionally, the power tool microcontroller **520** may transmit power tool device information (e.g., as described previously herein) to the external device **108** via the BLE microcontroller **535** and the BLE transceiver **540**.

[0039] FIG. **5**A illustrates a block diagram of a prior art power tool that includes the above-described components. As shown in FIG. **5**A, the power tool microcontroller **520**, the gate driver **525**, and the power manager **530** of the gate driver **525** are located within a system on chip (SOC) **550** (i.e., a single chip/integrated circuit located on, for example, a printed circuit board (PCB) within the power tool). Also as shown in FIG. **5**A, the BLE microcontroller **535** and BLE transceiver **540** are located within a wireless communication chip **560** (i.e., a BLE chip) that is separate from the SOC **550**. Separate chips **550** and **560** may be used within the power tool because some power tools may be manufactured without including the BLE chip **560** (e.g., see the block diagram of FIG. **5**B that includes the same components of FIG. **5**A with the exception of the BLE chip **560**).

[0040] However, having separate chips **550** and **560** for the power tool microcontroller **520** and the BLE microcontroller **535** and transceiver **540** may provide disadvantages within a power tool. For example, power tools may have limited space within their housing to accommodate components such as the components shown in FIGS. **5**A and **5**B. When two separate chips **550** and **560** are used, these separate chips take up more space inside the power tool. Furthermore, because the power tool of FIG. **5**A includes two separate microcontrollers **520** and **535**, these microcontrollers **520** and **535** include interface code to communicate with each other to transfer data. Along similar lines, the power tool includes wires and/or traces that couple the power tool microcontroller **520** to the BLE microcontroller **535** to allow for the microcontrollers **520** and **535** to communicate with each other. These wires and/or traces also consume space inside the power tool and provide sources of failure such as a risk of broken wires, damaged traces, and additional ingress risk.

[0041] To address and overcome the above-noted disadvantages, FIG. **6** illustrates a block diagram of the power tool **104** that includes a combined chip **675** that includes a microcontroller **620** that controls both power tool operation (e.g., motor control, light control, etc.) and communication between the power tool **104** and the external device **108**. The combined chip **675** may also include an integrated transceiver **680** (e.g., an integrated BLE transceiver and antenna) that allows the microcontroller **620** to bidirectionally communicate with the external device **108**. In some embodiments, the power tool **104** includes a transceiver and/or antenna that is separate from the combined chip **675** (i.e., that is not integrated into the combined chip **675**) and that is coupled to the combined chip **675** to allow the microcontroller **620** to bidirectionally communicate with the external device **108** via the transceiver. As used herein, the term chip refers to a monolithic integrated circuit, which includes electronic circuits integrated onto a single semiconductor material.

[0042] As shown in FIG. **6**, the power tool **104** may also include similar components as the power tool of FIG. **5**A (e.g., a motor **605**, a three-phase inverter **610** including six FETs, a battery pack **615**, a gate driver **625**, and a power manager **630** for the gate driver **625**). The chip design shown in FIG. **6** (in particular, the combined chip **675** that includes a single microcontroller **620** that controls both power tool operation (e.g., motor control, light control, etc.) and communication between the power tool **104** and the external device **108**) reduces or eliminates the above-noted disadvantages of the design shown in FIG. **5**A. For example, the combined chip **675** consumes less space inside the power tool **104** than the two separate chips **550** and **560** of FIG. **5**A. Additionally, the single microcontroller **620** may be configured to perform most or all of the functions that were previously performed by the two separate microcontrollers **520** and **535** of FIG. **5**A. Thus, the interface between the two separate microcontrollers **520** and **535** of FIG. **5**A (e.g., wires and/or traces) may be eliminated as well as any interface code that was previously required to allow for communication between the two separate microcontrollers **520** and **535**. Eliminating wires and/or traces may further reduce an amount of space inside the housing of the power tool **104** to, for example, allow the power tool **104** to be more compact and easier to maneuver and/or transport. Eliminating interface code may free up additional space in a memory of the power tool **104** such that additional usage data and/or configuration parameters could be stored, for example. Additionally, eliminating interface code may allow the power tool **104** to operate more efficiently (e.g., engage in bidirectional communication with the external device **108** more quickly) because less processing resources are necessary for such communication to occur. Furthermore, the inclusion of the combined chip **675** in the power tool **104** may simplify a manufacturing process of the power tool **104** by reducing programming steps during manufacturing that may be necessary to configure the two separate microcontrollers **520** and **535** of FIG. **5**A to be able to communicate with each other.

[0043] In some embodiments, as shown in FIG. **6**, the gate driver **625** and the power manager **630** are located within a driver chip **685** that is separate from the combined chip **675**, unlike the block diagram of the power tool shown in FIG. **5**A. While the driver chip **685** is separate from the combined chip **675** and may include additional wires and/or traces to couple the combined chip **675** with the driver chip **685**, the overall advantages described above that are gained with the combined tool control and BLE chip **675** outweigh any minor additional wires and/or traces that couple the combined chip **675** to the driver **685**. For example, the advantages provided by the reduction from the two microcontrollers **520** and **535** of FIG. **5**A to the single microcontroller **620** of FIG. **6** outweighs the additional wires and/or traces that couple the combined chip **675** to the driver **685**.

[0044] FIG. **7** illustrates another block diagram of the power tool **104** according to an alternative chip design. As shown in the embodiment of FIG. **7**, the power tool **104** may include similar components as the power tool of FIG. **6** (e.g., a motor **705**, a three-phase inverter **710** including six FETs, a battery pack **715**, a power tool microcontroller **720**, a gate driver **725**, a power manager **730** for the gate driver **725**, and a BLE chip **760** that includes a wireless communication microcontroller **735** (i.e., a BLE microcontroller) and a wireless communication transceiver **740** (i.e., a BLE transceiver)). As shown in FIG. **7**, the gate driver **725** and the power manager **730** of the gate driver **725** may be located on a first chip (i.e., a driver chip **785**), and the power tool microcontroller **720** may be located on a second chip (i.e., a power tool microcontroller chip) separate from the first chip. Additionally, the BLE chip **760** may be a third chip that is separate from each of the driver chip **785** and the power tool microcontroller chip. Because the microcontroller **720** is located on a separate chip from the driver chip **785**, different microcontrollers may be used for different power tool devices **102** as opposed to using the same microcontroller **520** that is integrated into the SOC **550** of FIG. **5**A for different power tool devices **102**.

[0045] FIG. **8** illustrates a further block diagram of the power tool **104** shown in FIG. **6**. As shown in FIG. **8**, the power tool **104** includes the motor **605**, the battery pack **615**, the gate driver **625**, and the combined chip **675**. In some embodiments, the combined chip **675** may include an electronic processor **805**, a memory **810**, and an antenna **815**. The power tool **104** may further include power switching elements **820**, Hall sensor(s) **825**, a battery pack interface **830**, and the actuator **430**. The components of the power tool **104** shown in FIG. **8** will be described in greater detail in the following paragraphs. In some embodiments, the microcontroller **620** of FIG. **6** includes the electronic processor **805** and the memory **810** of FIG. **8**. In some embodiments, the transceiver **680** of FIG. **6** includes the electronic processor **805** and the antenna **815** of FIG. **8**. In other words, the electronic processor **805**, the memory **810** and the antenna **815** may collectively embody the microcontroller **620** and the transceiver **680** of FIG. **6**. In some embodiments, the three-phase inverter **610** of FIG. **6** includes the power switching elements **820** of FIG. **8**.

[0046] In some embodiments, the motor **605** actuates or drives the drive device **425** (see FIG. **4**) and allows the drive device **425** to perform a particular task that the power tool **104** is configured to perform. The battery pack **615** (a primary power source) couples to the power tool **104** and provides electrical power to energize the motor **605**. The electronic processor **805** monitors a position of the actuator **430** and controls the motor **605** to be energized based on the position of the actuator **430**. Generally, when the actuator **430** is depressed, the motor **605** is energized, and when the actuator **430** is released, the motor **605** is de-energized. In the illustrated embodiment, the actuator **430** extends partially down a length of the handle **410** (see FIG. **4**); however, in other embodiments the actuator **430** extends down the entire length of the handle **410** or may be positioned elsewhere on the power tool **104**. The actuator **430** is moveably coupled to the handle **410** such that the actuator **430** moves with respect to the tool housing. Such movement is detectable by an electronic processor **805** of the power tool **104** (see FIG. **8**) through, for example, use of a Hall sensor, potentiometer, or the like. In some instances, a signal based on movement of the actuator **430** is binary and indicates either that the actuator **430** is depressed or released. In other instances, the signal indicates the position of the actuator **430** with more precision. For example, the signal may be an analog signal that varies from 0 to 5 volts depending on the extent that the actuator **430** is depressed. For example, 0 V output indicates that the actuator **430** is released, 1 V output indicates that the actuator 430 is 20% depressed, 2 V output indicates that the actuator **430** is 40% depressed, 3 V output indicates that the actuator **430** is 60% depressed 4 V output indicates that the actuator **430** is 80% depressed, and 5 V indicates that the actuator **430** is 100% depressed. The signal based on movement of the actuator **430** may be analog or digital.

[0047] In some embodiments, the battery pack interface **830** is coupled to the combined chip **675** and the battery pack **615**. The battery pack interface **830** includes a combination of mechanical (e.g., the battery pack receiving portion **415**) and electrical components configured to and operable for interfacing (e.g., mechanically, electrically, and communicatively connecting) the power tool **104** with the battery pack **615**. The battery pack interface **830** may include and/or be coupled to a power input unit (not shown). The battery pack interface **830** may transmit the power received from the battery pack **615** to the power input unit. The power input unit may include active and/or passive components (e.g., voltage step-down controllers, voltage converters, rectifiers, filters, etc.) to regulate or control the power received through the battery pack interface **830** and to the combined chip **675** and/or the motor **605**.

[0048] In some embodiments, the power switching elements **820** enable the electronic processor **805** of the combined chip **675** to control the operation of the motor **605** via the gate driver **625**. Generally, when the actuator **430** is depressed, electrical current is supplied from the battery pack interface **830** to the motor **605**, via the power switching elements **820**. When the actuator **430** is not depressed, electrical current is not supplied from the battery pack interface **830** to the motor **605**. In some embodiments, the amount that the actuator **430** is actuated is related to or corresponds to a desired speed of rotation of the motor **605**. In other embodiments, the amount that the actuator **430** is actuated is related to or corresponds to a desired torque.

[0049] In response to the electronic processor **805** determining that actuator **430** has been actuated, the electronic processor **805** provides a control signal to the gate driver **625** to activate the power switching elements **820** to provide power to the motor **605**. The power switching elements **820** control the amount of current available to the motor **605** and thereby control the speed and torque output of the motor **605**. The power switching elements **820** may include numerous FETs, bipolar transistors, or other types of electrical switches. For instance, the power switching elements **820** may include a six-FET bridge that receives pulse-width modulated (PWM) signals from the gate driver **625** to drive the motor **605** based on the control signal provided to the gate driver **625** from the electronic processor **805**.

[0050] In some embodiments, the power tool **104** includes sensors that are coupled to the electronic processor **805** and that communicate to the electronic processor **805** various signals indicative of different parameters of the power tool **104** or the motor **605**. The sensors may include Hall sensor(s) **825**, current sensor(s) (not shown), among other sensors, such as, for example, one or more voltage sensors, one or more temperature sensors, and one or more torque sensors. Each Hall sensor **825** outputs motor feedback information to the electronic processor **805**, such as an indication (e.g., a pulse) when a magnet of the rotor of the motor **605** rotates across the face of that Hall sensor. Based on the motor feedback information from the Hall sensors **825**, the electronic processor **805** can determine the position, velocity, and acceleration of the rotor. In response to the motor feedback information and the signals from sensor(s) indicating the position of the actuator **430**, the electronic processor **805** transmits control signals to the gate driver **625** to control the power switching elements **820** to drive the motor **605**. For instance, by selectively enabling and disabling the power switching elements **820**, power received via the battery pack interface **830** is selectively applied to stator coils of the motor **605** to cause rotation of its rotor. The motor feedback information is used by the electronic processor **805** and/or the gate driver **625** to ensure proper timing of control signals to the power switching elements **820** and, in some instances, to provide closed-loop feedback to control the speed of the motor **605** to be at a desired level.

[0051] As a more particular example, to drive the motor **605**, the electronic processor **805** (via the gate driver **625**) enables a first high side FET and first low side FET pair (e.g., by providing a voltage at a gate terminal of the FETs) for a first period of time. In response to determining that the rotor of the motor **605** has rotated based on a pulse from the Hall sensors **825**, the electronic processor **805** (via the gate driver **625**) disables the first FET pair, and enables a second high side FET and a second low side FET. In response to determining that the rotor of the motor **605** has rotated based on pulse(s) from the Hall sensors **825**, the electronic processor **805** (via the gate driver **625**) disables the second FET pair, and enables a third high side FET and a third low side FET. This sequence of cyclically enabling pairs of high side and low side FETs repeats to drive the motor **605**. Further, in some embodiments, one or both of the control signals to each FET pair includes pulse width modulated (PWM) signals having a duty cycle that is set in proportion to the amount of trigger pull to thereby control the speed or torque of the motor **605**.
[0052] As shown in FIG. **8**, the combined chip **675**, and in particular the electronic processor **805**, is electrically and/or communicatively connected to a variety of components of the power tool **104**. In some embodiments, the combined chip **675** includes a plurality of electrical and electronic components that provide power, operational control, and protection to the components within the combined chip **675** and/or power tool **104**. For example, the combined chip **675** includes, among other things, the electronic processor **805** (e.g., a microprocessor, a microcontroller, or another suitable programmable device), the memory **810**, and input/output units (e.g., input/output pins). The electronic processor **805** may include, among other things, a control unit, an arithmetic logic unit ("ALU"), and a plurality of registers. In some embodiments, the combined chip **675** is implemented partially or entirely on a semiconductor (e.g., a field-programmable gate array ["FPGA"] semiconductor) chip, such as a chip developed through a register transfer level ("RTL") design process. Although FIG. **8** shows the combined chip **675** as including a single electronic processor **805**, in some embodiments, the combined chip **675** includes additional electronic processors. For example, the combined chip **675** may include a first electronic processor configured to control operation of the motor **605** as described above and a second electronic processor configured to manage wireless communication to and from the power tool **104** via the antenna **815**.
[0053] The memory **810** includes, for example, a program storage area and a data storage area. The memory **810** may include combinations of different types of memory, such as read-only memory ("ROM"), random access memory ("RAM") (e.g., dynamic RAM ["DRAM"], synchronous DRAM ["SDRAM"], etc.), electrically erasable programmable read-only memory ("EEPROM"), flash memory, a hard disk, an SD card, or other suitable magnetic, optical, physical, or electronic memory devices. The electronic processor **805** is connected to the memory **810** and executes software instructions that are capable of being stored in a RAM of the memory **810** (e.g., during execution), a ROM of the memory **810** (e.g., on a generally permanent basis), or another non-transitory computer readable medium such as another memory or a disc. Software included in the implementation of the power tool **104** can be stored in the memory **810**. The software includes, for example, firmware, one or more applications, program data, filters, rules, one or more program modules, and other executable instructions. The electronic processor **805** is configured to retrieve from the memory **810** and execute, among other things, instructions related to the control processes and methods described herein. The electronic processor **805** is also configured to store power tool device information on the memory **810** including operational data, information identifying the type of tool, a unique identifier for the particular tool, and other information relevant to operating or maintaining the power tool **104**. The power tool device information, such as current levels, motor speed, motor acceleration, motor direction, number of impacts, may be captured or inferred from data output by the sensors included in the power tool **104**. Such power tool device information may then be accessed by a user with the external device **108**. In other constructions, the combined chip **675** includes additional, fewer, or different components. For example, the gate driver **625** or

functionality implemented by the gate driver **625** may be included and/or implemented within the combined chip **675** rather than being included in a second chip that is separate from the combined chip **675** as shown in FIG. **6**.

[0054] In some embodiments, the combined chip **675**, and in particular the electronic processor **805**, also acts as a wireless communication controller. As shown in FIG. **8**, the combined chip **675** includes an integrated antenna **815**. The combined chip **675** may also include a radio transceiver coupled to the antenna **815** and to the electronic processor **805** to allow the electronic processor **805** to bidirectionally communicate with the external device **108** via the antenna **815**. In some embodiments, the antenna **815** may not be integrated with the combined chip **675** and may be located elsewhere in the power tool **104**. However, in such embodiments, the combined chip **675** may still act as the wireless communication controller such that a separate wireless communication chip is not used within the power tool **104** as explained previously herein. For example, the electronic processor **805** may nevertheless act as a wireless transceiver to bidirectionally communicate with the external device **108** via an antenna that is not integrated into the combined chip **675**.

[0055] In some embodiments, the memory **810** can store instructions to be implemented by the electronic processor **805** and/or may store data related to communications between the power tool **104** and the external device **108** or the like. The electronic processor **805** controls wireless communications between the power tool **104** and the external device **108**. For example, the electronic processor **805** buffers incoming and/or outgoing data and determines the communication protocol and/or settings to use in wireless communications.

[0056] In the illustrated embodiment, the electronic processor **805** may include a Bluetooth® controller. The Bluetooth® controller communicates with the external device **108** employing the Bluetooth® protocol. Therefore, in the illustrated embodiment, the external device **108** and the power tool **104** are within a communication range (i.e., in proximity) of each other while they exchange data. In other embodiments, the electronic processor **805** communicates using other protocols (e.g., Wi-Fi, cellular protocols, a proprietary protocol, etc.) over a different type of wireless network. For example, the electronic processor **805** may be configured to communicate via Wi-Fi through a wide area network such as the Internet or a local area network, or to communicate through a piconet (e.g., using infrared or NFC communications). The communication between the power tool **104** and the external device **108** may be encrypted to protect the data exchanged between the power tool **104** and the external device/network **108** from third parties.

[0057] The electronic processor **805** may periodically broadcast an advertisement message that may be received by an external device **108** in communication range of the power tool **104**. The advertisement message may include identification information regarding the tool identity, remaining capacity of a battery pack **615** attached to the power tool **104**, and other limited amount of power tool device information. The advertisement message may also identify the product as being from a particular manufacturer or brand.

[0058] When the power tool **104** and the external device **108** are within communication range of each other, the electronic processor **805** may establish a communication link (e.g., pair) with the external device **108** to allow the external device **108** to obtain and export power tool device information such as tool usage data, maintenance data, mode information, drive device information, and the like from the power tool **104**. The exported information can be used by tool users or owners to log data related to a particular power tool **104** or to specific job activities. The exported and logged data can indicate when work was accomplished and that work was accomplished to specification. The logged data can also provide a chronological record of work that was performed, track duration of tool usage, and the like. While paired with the external device **108**, the electronic processor **805** may also import (i.e., receive) information from the external device **108** into the power tool **104** such as, for example, configuration information such as operation thresholds, maintenance thresholds, mode configurations, programming for the power tool **104**, software updates, and the like.

[0059] In some embodiments, the power tool **104** may include fewer or additional components in configurations different from that illustrated in FIG. **8**. For example, in some embodiments, the power tool **104** includes indicators (e.g., light-emitting diodes (LEDs) and/or a display screen) that are coupled to the electronic processor **805** and receive control signals from the electronic processor **805** to turn on and off or otherwise convey information based on different states of the power tool **104**. For example, the indicators may be configured to indicate measured electrical characteristics of the power tool **104**, the status of the power tool **104**, the mode of the power tool **104**, and the like. The indicators may also include elements to convey information to a user through audible or tactile outputs. As another example, the power tool **104** may include a real-time clock (RTC) configured to increment and keep time independently of the other power tool components. Having the RTC as an independently powered clock enables time stamping of operational data (stored in memory **810** for later export) and a security feature whereby a lockout time is set by a user and the tool is locked-out when the time of the RTC exceeds the set lockout time. As another example, the power tool **104** may include a location component (for example, a global positioning system receiver) used for tracking a location of the power tool **104**. As another example, the power tool **104** may not include Hall sensor(s) **825** to monitor rotational position information of the motor **605**. Rather, the power tool **104** may implement a sensor-less design to monitor rotational position of the motor **605**, for example, by monitoring back electromotive force (EMF) of the motor **605**.

[0060] In some embodiments, the power tool **104** includes one or more printed circuit boards (PCBs) that include one or more of the electrical components shown in FIG. **8**. FIGS. **9**A-C illustrate example locations within the power tool **104** where the PCBs may be positioned. As shown in FIGS. **9**A-C, in some embodiments, the power tool **104** includes a Hall sensor PCB located at position **905** in front of the motor **605**. In other embodiments, the Hall sensor PCB may be located behind the motor **605** or the Hall sensor PCB may not be present within the power tool **104**. FIGS. **9**A, **9**B, and **9**C illustrate board locations **910**, **915**, and **920**, respectively, which are locations at which a control PCB **1005** (shown in FIG. **10**) may be located within the power tool **104**. For example, as shown in FIG. **9**A, in some embodiments, a control PCB (e.g., the control PCB **1005** of FIG. **10**) that includes the combined chip **675** is located in the handle **410** of the power tool **104** at the location **910**. As shown in FIG. **9**B, in some embodiments, a control PCB (e.g., the control PCB **1005**) that includes the combined chip **675** is located above the actuator **430** and the handle **410**, but below the motor **605** and drive device **425**, at the location **915**. As shown in FIG. **9**C, in some embodiments, a control PCB (e.g., the control PCB **1005**) that includes the combined chip **675** is located below the handle **410** and above the battery pack receiving portion **415**, at the location **920**. In some embodiments, the control PCB at the locations **910**, **915**, and **920** (e.g., the control PCB **1005**) also includes at least one of the gate driver **625** and the power switching elements **820**, in addition to the combined chip **675**. In some embodiments, the components explained above as being included on the control PCB **1005** may instead be located on the Hall sensor PCB.

[0061] FIG. **10** illustrates the control PCB **1005** according to one example embodiment. In some embodiments, the control PCB **1005** is approximately **70** millimeters long and **29** millimeters wide. In some embodiments, a controller portion **1010** of the control PCB **1005** includes the combined chip **675** and is approximately **30** millimeters long and **26** millimeters wide. As shown in FIG. **10**, in some embodiments, the controller portion **1010** may include approximately half of the control PCB **1005** or may include less than half of the control PCB **1005**. As mentioned above, in some embodiments, the control PCB **1005** includes the gate driver **625** and the power switching elements **820** in addition to the combined chip **675**. The term “approximately” is defined as being close to as understood by one of ordinary skill in the art, and in one non-limiting embodiment the term is defined to be within 10%, in another embodiment within 5%, in another embodiment within 1% and in another embodiment within 0.5%. The size, shape, and location of components on the

control PCB **1005** shown in FIG. **10** is an example. In some embodiments, the size, shape, and location of components on the control PCB **1005** may be different.
[0062] FIG. **11** illustrates a flowchart of a method **1100** performed by the electronic processor **805** of the combined chip **675** according to one example embodiment. While a particular order of processing steps, message receptions, and/or message transmissions is indicated in FIG. **11** as an example, timing and ordering of such steps, receptions, and transmissions may vary where appropriate without negating the purpose and advantages of the examples set forth in detail throughout the remainder of this disclosure.
[0063] At block **1105**, the electronic processor **805** of the combined chip **675** receives configuration information from the external device **108** via the integrated antenna **815**. At block **1110**, the electronic processor **805** determines that the actuator **430** has been actuated. At block **1115**, in response to determining that the actuator **430** has been actuated, the electronic processor **805** provides a signal to the gate driver **625**, and the gate driver **625** is configured to control the power switching elements **820** based on the signal. In some embodiments, the signal is generated at least in part based on the configuration information received (e.g., which may include motor control parameters specified by a user via a control screen on the external device **108**). At block **1120**, the electronic processor **805** receives the motor positional information from the Hall sensor(s) **825**. At block **1125**, the electronic processor **805** controls the signal provided to the gate driver **625** based on the motor positional information and based on the configuration information received from the external device **108**. At block **1130**, the electronic processor **805** transmits power tool device information to an external device **108** via the integrated antenna **815**. As indicated in FIG. **11**, the electronic processor **805** proceeds back to block **1105** after completing block **1130** to repeat one or more blocks of the method **1100** (e.g., to continue operating the motor **605** in accordance with the actuation of the actuator **430** and the received motor positional information). Example techniques to implement each of the blocks in the method **1100** are provided in further detail in the preceding discussion with respect to FIGS. **1-8**.
[0064] As explained previously herein, the above description of the combined chip **675** may be implemented in other power tool devices such as work lights, battery packs, and the like.
[0065] Thus, embodiments described herein provide, among other things, a power tool that communicates with an external device for configuring the power tool and obtaining data from the power tool.

Claims

1. A power tool device comprising: an actuator configured to be actuated by a user; a powered element; a power switching element configured to control whether power is provided to the powered element; a gate driver coupled to the power switching element and configured to control the power switching element; a printed circuit board (PCB); a wireless transceiver; and a combined power tool device control and wireless communication control chip located on the PCB and coupled to the actuator and the gate driver, wherein the combined power tool device control and wireless communication control chip includes a memory and one or more electronic processors, the one or more electronic processors configured to control both (i) power tool device operation and (ii) wireless communication between the power tool device and an external device, wherein the one or more electronic processors of the combined power tool device control and wireless communication control chip control the wireless communication between the power tool device and the external device (i) by interfacing directly with the wireless transceiver and (ii) without interfacing with any intermediate electrical components of the power tool device that are external to the combined power tool device control and wireless communication control chip between the combined power tool device control and wireless communication control chip and the wireless transceiver.

2. The power tool device of claim 1, wherein the powered element includes a work light, and wherein the one or more electronic processors is configured to control power tool device operation by controlling a signal provided to the gate driver to control a work light duration, a work light intensity, or both the work light duration and the work light intensity.
3. The power tool device of claim 1, wherein the power tool device includes a work light, and wherein a primary function of the power tool device includes lighting an area.
4. The power tool device of claim 1, wherein the powered element includes a motor, and further comprising a Hall effect sensor configured to monitor motor position information of the motor; wherein the combined power tool device control and wireless communication control chip is coupled to the Hall effect sensor; and wherein the one or more electronic processors is configured to receive the motor position information from the Hall effect sensor, and control a signal provided to the gate driver based on the motor position information.
5. The power tool device of claim 1, further comprising a power manager configured to: monitor a characteristic of the powered element, the power switching element, or both the powered element and the power switching element; and control the gate driver based on the characteristic that is monitored.
6. The power tool device of claim 1, wherein the power switching element is located on the PCB.
7. The power tool device of claim 1, wherein the one or more electronic processors is configured to: wirelessly transmit power tool device information to the external device, wherein the power tool device information includes at least one of the group consisting of usage data, maintenance data, mode information, and combinations thereof; and wirelessly receive configuration information from the external device, wherein the configuration information includes at least one of the group consisting of a motor speed parameter, a motor torque parameter, a work light parameter, and combinations thereof.
8. The power tool device of claim 1, wherein the wireless transceiver is integrated into the combined power tool device control and wireless communication control chip, wherein the combined power tool device control and wireless communication control chip includes an integrated antenna, and wherein the one or more electronic processors is configured to: wirelessly transmit power tool device information to the external device via the integrated antenna; and wirelessly receive configuration information from the external device via the integrated antenna, wherein the one or more electronic processors is configured to use the configuration information to control a signal that is provided to the gate driver.
9. A method of operating a power tool device, the method comprising: controlling, with one or more electronic processors of the power tool device, both (i) power tool device operation and (ii) wireless communication between the power tool device and an external device, wherein the one or more electronic processors is included in a combined power tool device control and wireless communication control chip that includes a memory, the combined power tool device control and wireless communication control chip being located on a printed circuit board (PCB) and coupled to an actuator of the power tool device, and wherein the one or more electronic processors of the combined power tool device control and wireless communication control chip control the wireless communication between the power tool device and the external device (i) by interfacing directly with a wireless transceiver and (ii) without interfacing with any intermediate electrical components of the power tool device that are external to the combined power tool device control and wireless communication control chip between the combined power tool device control and wireless communication control chip and the wireless transceiver; and controlling, with a gate driver of the power tool device, a power switching element configured to control whether power is provided to a powered element of the power tool device.
10. The method of claim 9, wherein the powered element includes a work light, and further comprising providing, with the one or more electronic processors, a signal to the gate driver to control a work light duration, a work light intensity, or both the work light duration and the work

light intensity.
11. The method of claim 9, wherein the power tool device includes a work light, and wherein a primary function of the power tool device includes lighting an area.
12. The method of claim 9, wherein the powered element includes a motor, and further comprising: receiving, with the one or more electronic processors, motor position information of the motor from a Hall effect sensor coupled to the combined power tool device control and wireless communication control chip; and controlling, with the one or more electronic processors, a signal provided to the gate driver based on the motor position information.
13. The method of claim 9, further comprising: monitoring, with a power manager, a characteristic of the powered element, the power switching element, or both the powered element and the power switching element; and controlling, with the power manager, the gate driver based on the characteristic that is monitored.
14. The method of claim 9, wherein the power switching element is located on the PCB.
15. The method of claim 9, further comprising: wirelessly transmitting, with the one or more electronic processors, power tool device information to an external device, wherein the power tool device information includes at least one of the group consisting of usage data, maintenance data, mode information, and combinations thereof; and wirelessly receiving, with the one or more electronic processors, configuration information from the external device, wherein the configuration information includes at least one of the group consisting of a motor speed parameter, a motor torque parameter, a work light parameter, and combinations thereof.
16. The method of claim 9, wherein the wireless transceiver is integrated into the combined power tool device control and wireless communication control chip, and further comprising: wirelessly transmitting, with the one or more electronic processors, power tool device information to an external device via an integrated antenna included in the combined power tool device control and wireless communication control chip; and wirelessly receiving, with the one or more electronic processors, configuration information from the external device via the integrated antenna.
17. A power tool device comprising: an actuator configured to be actuated by a user; a powered element; a power switching element configured to control whether power is provided to the powered element; a gate driver coupled to the power switching element and configured to control the power switching element; a printed circuit board (PCB); an antenna; and a combined power tool device control and wireless communication control chip located on the PCB and coupled to the actuator, the gate driver, and the antenna, wherein the combined power tool device control and wireless communication control chip includes a memory and one or more electronic processors, the one or more electronic processors configured to control both (i) power tool device operation and (ii) wireless communication between the power tool device and an external device, wherein the one or more electronic processors of the combined power tool device control and wireless communication control chip control the wireless communication between the power tool device and the external device (i) by interfacing directly with a wireless transceiver that is coupled to the antenna and (ii) without interfacing with any intermediate electrical components of the power tool device that are external to the combined power tool device control and wireless communication control chip between the combined power tool device control and wireless communication control chip and the wireless transceiver.
18. The power tool device of claim 17, wherein the powered element includes a work light, and wherein the one or more electronic processors is configured to control a signal provided to the gate driver to control a work light duration, a work light intensity, or both the work light duration and the work light intensity.
19. The power tool device of claim 17, wherein the power tool device includes a work light, and wherein a primary function of the power tool device includes lighting an area.
20. The power tool device of claim 17, wherein the powered element includes a motor, and further comprising a Hall effect sensor configured to monitor motor position information of the motor;

wherein the combined power tool device control and wireless communication control chip is coupled to the Hall effect sensor; and wherein the one or more electronic processors is configured to receive the motor position information from the Hall effect sensor, and control a signal provided to the gate driver based on the motor position information.